

ADDITIONAL PROBLEMS

1. The number of integers in the range of

$$f(x) = \frac{1}{\pi} (\sin^{-1}(x) + \tan^{-1}(x)) + \frac{x+1}{x^2+2x+5}$$

is

- a) 0 b) 3 c) 2 d) 1

Key. c

Sol.
$$f(x) = \frac{1}{\pi} (\sin^{-1}(x) + \tan^{-1}(x)) + \frac{1}{(x+1) + \frac{4}{(x+1)}}$$

$$= g(x) + h(x)$$

Here Domain of $g(x)$ is $[-1, 1]$

$$\Rightarrow \text{range of } g(x) \text{ is } \left[\frac{-3}{4}, \frac{3}{4} \right]$$

Also, max value of $h(x)$ occurs when

$$(x+1) + \frac{4}{(x+1)} \text{ is min. at } x = 1$$

$$\Rightarrow \text{range of } f(x) \text{ is } \left[\frac{-3}{4}, 1 \right]$$

$\Rightarrow$ Number of integers = 2

2. If $f(x)$ is continuous function and attains only non-zero rational values, then the number of distinct roots of the equation

$$f(0).x^2 - 6f(3)x + 9f(-5) = 0 \text{ is}$$

- a) 1 b) 2 c) 0 d) ∞

Key. a

Sol. As function attains only rational values, so $f(x)$ is constant function

$$\Rightarrow f(0) = f(3) = f(-5) \neq 0$$

$$\Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x-3)^2 = 0$$

$$\Rightarrow x = 3$$

3. the number of functions defined from $f: \{1, 2, 3, 4, 5, 6\} \rightarrow \{7, 8, 9, 10\}$ such that

$$\sum_{r=1}^6 f(r) \text{ is odd, is}$$

- a) 2^{10} b) 2^{11} c) 2^{12} d) $2^{12} - 1$

Key. b

Sol. Sum can be odd if one (or) three (or) five elements link with odd number

$$\therefore {}^6C_1 \cdot 2 \cdot 2^5 + {}^6C_3 \cdot 2^3 \cdot 2^3 + {}^6C_5 \cdot 2^5 \cdot 2$$

$$= 2^6 ({}^6C_1 + {}^6C_3 + {}^6C_5) = 2^6 \cdot 2^5 = 2^{11}$$

4. If $f(x) = \sin^{-1}(x)$ and

$$g(x) = [\sin(\cos x)] + [\cos(\sin x)] \text{ then the}$$

range of $f(g(x))$ is, when $[\cdot]$ is G.I.F

- a) $\left\{ -\frac{\pi}{2}, \frac{\pi}{2} \right\}$ b) $\left\{ -\frac{\pi}{2}, 0 \right\}$
c) $\left\{ 0, \frac{\pi}{2} \right\}$ d) $\left\{ -\frac{\pi}{2}, 0, \frac{\pi}{2} \right\}$

Key. d

Sol. $\cos x \in [-1, 1] \Rightarrow \sin(\cos x) \in [-\sin 1, \sin 1]$

$$\Rightarrow [\sin(\cos x)] = 0, -1$$

$$\text{Similarly } [\cos(\sin x)] = 0, 1$$

$$g(x) = -1, 0, 1$$

$$\text{so, range of } f(g(x)) \text{ is } \sin^{-1}(-1), \sin^{-1}(0), \sin^{-1}(1)$$

$$= \left\{ -\frac{\pi}{2}, 0, \frac{\pi}{2} \right\}$$

5. Let $f: R - \{0\} \rightarrow [-1, 1]$ defined by

$$f(x) = \frac{e^{x^3} + e^{-x^3}}{e^{x^3} - e^{-x^3}}, \text{ then 'f' is}$$

- a) Many-one function b) one-one function
c) decreasing in $(-\infty, 0)$

d) decreasing in $(0, \infty)$

Key. b, c, d

Sol. $f(x)$ is not defined at $x = 0$

$$f'(x) = \frac{-12x^2}{(e^{x^3} - e^{-x^3})^2} < 0, x \neq 0$$

6. If $f: R \rightarrow \left[\frac{\pi}{6}, \frac{\pi}{2}\right)$ defined by

$$f(x) = \sin^{-1}\left(\frac{x^2 - a}{x^2 + 1}\right) \text{ is an onto function,}$$

then the set of values of 'a' is

a) $(-\infty, -1)$ b) $(-1, \infty)$

c) $(-\infty, 0)$ d) $(-\infty, \infty)$

Key. b

Sol. Since $f(x)$ is onto $\frac{\pi}{6} \leq \sin^{-1}\left(\frac{x^2 - a}{x^2 + 1}\right) < \frac{\pi}{2}$

$$\Rightarrow \frac{1}{2} \leq \frac{x^2 - a}{x^2 + 1} < 1$$

$$\Rightarrow \frac{1}{2} \leq 1 - \frac{(a+1)}{x^2 + 1} < 1, \forall x \in R$$

$$\Rightarrow a+1 > 0 \Rightarrow a \in (-1, \infty)$$

Passage: Let $f(x) = x^2 - 2x - 1, \forall x \in R$. Let $f: (-\infty, a] \rightarrow [b, \infty)$, where 'a' is the largest real number for which $f(x)$ is bijective. Now answer the following.

7. Let $f: R \rightarrow R; g(x) = f(x) + 3x - 1$ then the least value of the function $y = g(|x|)$ is

a) $-\frac{9}{4}$ b) $-\frac{5}{4}$ c) -2 d) -1

Key. c

8. Let $f: [a, \infty) \rightarrow [b, \infty)$ then $f^{-1}(x)$ is given by

a) $1 + \sqrt{x+2}$ b) $1 - \sqrt{x+3}$

c) $1 - \sqrt{x+2}$ d) $1 + \sqrt{x+3}$

Key. a

9. Let $f: R \rightarrow R$, then the range of values of 'K' for which the equation $f(|x|) = K$ has four distinct real roots, is

a) $(-2, -1)$ b) $(-2, 0)$

c) $(-1, 0)$ d) $(0, 1)$

Key. a

Sol. $f(x) = (x-1)^2 - 2 \Rightarrow a = 1; b = -2$

$$g(x) = x^2 + x - 2 = \left(x + \frac{1}{2}\right)^2 - \frac{9}{4}$$

$$\Rightarrow g(|x|) = \left(|x| + \frac{1}{2}\right)^2 - \frac{9}{4}$$

$$g_{\min} = g(0) = -2$$

$$f: [1, \infty) \rightarrow [-2, \infty)$$

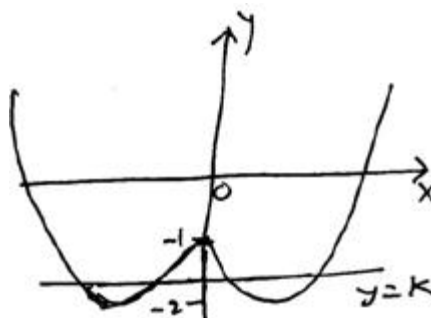
$$f^{-1}: [-2, \infty) \rightarrow [1, \infty)$$

$$f(x) = y \Rightarrow x^2 - 2x - (1+y) = 0$$

$$\Rightarrow x = 1 \pm \sqrt{2+y} \Rightarrow f^{-1}(y) = 1 + \sqrt{2+y}$$

$$f^{-1}(x) = 1 + \sqrt{2+x}$$

graph of $y = f(|x|)$



from the graph,

$f(|x|) = K$ has four roots if $K \in (-2, -1)$

10. If a function $f: R \rightarrow R$ be such that

$$f(x - f(y)) = f(f(y)) + x.f(y) + f(x) - 1,$$

$\forall x, y \in R$ then $f(2) =$

- a) 1 b) 3 c) -1 d) 4

Key. c

Sol. Put $x = f(y) = 0$

$$\text{then } f(0) = f(0) + 0 + f(0) - 1$$

$$\Rightarrow f(\lambda) = 1 - \frac{\lambda^2}{2} \Rightarrow f(x) = 1 - \frac{x^2}{2}$$

11. If fractional part of $\frac{1}{x}$ and x^2 for some

$x \in (\sqrt{2}, \sqrt{3})$ are equal, then the value of

$$x^4 - \frac{3}{x} \text{ is}$$

- a) 3 b) 5 c) 7 d) 0

Key. b

$$\text{Sol. } \left\{ \frac{1}{x} \right\} = \{x^2\} \Rightarrow \frac{1}{x} = x^2 - 2$$

$$\Rightarrow x^3 - 2x - 1 = 0 \Rightarrow (x+1)(x^2 - x - 1) = 0$$

where $x = -1 \notin (\sqrt{2}, \sqrt{3})$

$$\Rightarrow x^2 - x - 1 = 0 \Rightarrow x^4 = 3x + 2$$

$$\Rightarrow x^4 - \frac{3}{x} = (3x + 2) - (x^2 - 2)$$

12. If the domain of $f(x)$ is $[1, 3]$ then the

domain of $f(\log_2(x^2 + 3x - 2))$

a) $[-5, -4] \cup [1, 2]$ b) $[-13, -2] \cup \left[\frac{3}{5}, 5\right]$

c) $[4, 10] \cup [2, 7]$ d) $[-3, 2]$

Key. a

$$\text{Sol. } 1 \leq \log_2(x^2 + 3x - 2) \leq 3$$

$$2 \leq x^2 + 3x - 2 \leq 8$$

$$-5 \leq x \leq -4 \text{ and } 1 \leq x \leq 2$$

13. Let 'g' be a function defined by

$$g(x) = \sqrt{\{x\}} + \sqrt{1 - \{x\}} \text{ where } \{.\} \text{ denotes}$$

the fraction part function, then g is

- a) odd function b) even function
c) one-one function d) many-one function

Key. b, d

Sol. If $x \in Z$ then $\{x\} = 0$

$$\Rightarrow g(x) = 1$$

$$\Rightarrow g(-x) = g(x) = 1$$

If $x \notin Z$ then $\{-x\} = 1 - \{x\}$

$$\begin{aligned} \therefore g(-x) &= \sqrt{\{-x\}} + \sqrt{1 - \{-x\}} \\ &= \sqrt{1 - \{x\}} + \sqrt{1 - (1 - \{x\})} \\ &= \sqrt{1 - \{x\}} + \sqrt{\{x\}} = g(x) \end{aligned}$$

14. Consider $f: R^+ \rightarrow R$ such that $f(3) = 1$ for

$a \in R^+$ and

$$f(x) \cdot f(y) + f\left(\frac{3}{x}\right) \cdot f\left(\frac{3}{y}\right) = 2f(xy),$$

$\forall x, y \in R^+$, then $f(97) =$

- a) 1 b) -1 c) 2 d) 97

Key. a

Sol. Put $x = y = 1 \Rightarrow f(1) = 1$

$$\text{put } y = 1 \Rightarrow f(x) = f\left(\frac{3}{x}\right), \forall x > 0$$

$$f(x) \cdot f\left(\frac{3}{x}\right) + f\left(\frac{3}{x}\right) \cdot f(x) = 2f(3)$$

$$\Rightarrow f(x) \cdot f\left(\frac{3}{x}\right) = 1 \Rightarrow f(x) = 1 \quad (\because x > 0)$$

15. If the solution set for $f(x) < 3$ is $(0, \infty)$ and the solution set for $f(x) > -2$ is $(-\infty, 5)$ then the interval in which 'x' lies for the equation $(f(x))^2 \geq f(x) + 6$, is

- a) $(-\infty, \infty)$ b) $(-\infty, -5] \cup [5, \infty)$
c) $(-\infty, 0] \cup [5, \infty)$ d) $[-2, 5]$

Key. c

Sol. $f^2(x) - f(x) - 6 \geq 0$
 $\Rightarrow (f(x) - 3)(f(x) + 2) \geq 0$
 $\Rightarrow f(x) \in (-\infty, -2] \cup [3, \infty)$
 $x \in (-\infty, 0] \cup [5, \infty)$

16. Number of integers in the domain of

$f(x) = \log_{[x^2]}(4 - |x|) + \log_2\{\sqrt{x}\}$ is (where [.] and { } have G.I.F, fraction part)

- a) 0 b) 1 c) 2 d) 3

Key. c

Sol. $4 - |x| > 0 \Rightarrow x \in (-4, 4)$
 $[x^2] > 0; [x^2] \neq 1 \Rightarrow [x^2] \geq 2$
 $x \in (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty)$
 $\{\sqrt{x}\} > 0; \sqrt{x} \notin I \Rightarrow x = 2, 3$

17. Let $f(x) = ax^7 + bx^3 + cx - 5$, where a, b, c are constants, if $f(-7) = 7$ then $f(7) =$

- a) -17 b) -7 c) 14 d) 7

Key. a

Sol. $f(-x) + f(x) = -10$
 $\Rightarrow f(-7) + f(7) = -10$

18. If $f(x)$ is a polynomial having integral coefficients such that $f(m) - f(n) = 1$, where 'm' and 'n' are integers, then the value of $m - n$, is

- a) 2 b) -2 c) 1 d) -1

Key. c, d

Sol. Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_kx^k$
 (where $a_0, a_1, a_2, \dots$ are integers)

$$f(m) - f(n) = 1$$

$$\Rightarrow a_1(m - n) + a_2(m^2 - n^2) + \dots$$

$$a_k(m^k - n^k) = 1$$

$$\Rightarrow (m - n) (\text{integers}) = 1$$

$$\Rightarrow (m - n) \text{ can be } 1 \text{ (or) } -1$$

according as Integer = 1, -1

19. If $f(x) = kx^2 - m$ such that $f(1) \in [-4, -1]$ and $f(2) \in [-1, 5]$ then the max. value of $f(3)$ is

- a) 10 b) 15 c) 17 d) 20

Key. d

Sol. $f(1) = k - m$

$$f(2) = 4k - m$$

$$\Rightarrow k = \frac{f(2) - f(1)}{3}; m = \frac{f(2) - 4f(1)}{3}$$

$$\Rightarrow f(3) = 9k - m = \frac{1}{3}(8f(2) - 5f(1))$$

$$\Rightarrow f(3) \in [-1, 20]$$

20. The number of pairs of integers (x, y) that satisfy $\cos(xy) = x; \tan(xy) = y$ is

- a) 1 b) 2 c) 4 d) 6

Key. a

Sol. $\frac{\sin(xy)}{\cos(xy)} = y \Rightarrow \sin(xy) = xy \Rightarrow xy = 0$

$$\Rightarrow y = 0 (\because x \neq 0) \Rightarrow x = 1$$

FUNCTIONS

21. Consider a polynomial $f(x)$ which satisfies

$$f(x) = (f'(x))^2; \forall x \in R \text{ then } f(x) \text{ can be}$$

- a) linear function b) quadratic function
c) a cubic function
d) any polynomial of even degree

Key. b

Sol. $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

on differentiating $f'(x)$ will be of $(n-1)^{th}$ degree

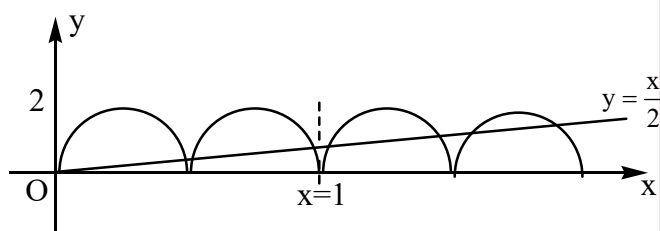
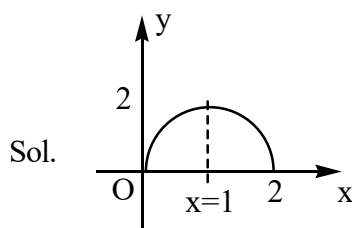
$$\text{equating degrees} \Rightarrow n = 2(n-1) \Rightarrow n = 2$$

22.. Let $f(x) = 2x(2-x); 0 \leq x \leq 2$, then the

number of solutions of $f(f(f(x))) = \frac{x}{2}$ is

- a) 2 b) 4 c) 8 d) 12

Key. c



Since the graph $y = f(x)$ is symmetrical about the line $x = 1$

Graph of $y = f(f(x))$ and

$y = f(f(f(x)))$ are also symmetrical about the line $x = 1$

Number of solutions = 8

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23. Total number of values of 'x' of the form

$$\frac{1}{n}, n \in N \text{ in the interval } x \in \left[\frac{1}{25}, \frac{1}{10} \right] \text{ which}$$

satisfy the equation

$$\{x\} + \{2x\} + \dots + \{12x\} = 78x \text{ (where } \{.\} \text{ is fractional part function)}$$

- a) 12 b) 13 c) 14 d) 15

Key. c

Sol. $(x - [x]) + (2x - [2x]) + \dots + (12x - [12x]) = 78x$

$$\Rightarrow [x] + [2x] + \dots + [12x] = 0$$

$$\Rightarrow 0 \leq x < \frac{1}{12} \Rightarrow \frac{1}{25} \leq x \leq \frac{1}{10}$$

$$\therefore \frac{1}{25} \leq x \leq \frac{1}{12}$$

24. Number of integers in the range of

$$f(x) = \frac{1}{\pi} (\sin^{-1} x + \tan^{-1} x) + \frac{x+1}{x^2+2x+5} \text{ is}$$

- a) 0 b) 3 c) 2 d) 1

Sol. Let $f(x) = g(x) + h(x)$ where domain of $g(x)$ is $[-1, 1]$

$$\therefore \text{Max. of } g(x) = g(1) = \frac{3}{4} \text{ and min. of}$$

$$g(x) = g(-1) = -\frac{3}{4}$$

Also max. of $h(x)$ occurs when $x+1 + \frac{4}{x+1}$ is minimum at $x = 1$

$$\therefore \text{range of } f(x) = \left[-\frac{3}{4}, 1 \right]$$

25. Consider $f: R^+ \rightarrow R$ such that $f(3) = 1$ for

$$R^+ \text{ and } f(x) \cdot f(y) + f\left(\frac{3}{x}\right) f\left(\frac{3}{y}\right)$$

$$= 2f(xy), \forall x, y \in R^+ \text{ then } f(97) =$$

Sol. Put $x = y = 1 \Rightarrow f(1) = 1$

Now put $y = 1$

$$\Rightarrow f(x) = f\left(\frac{3}{x}\right), \forall x > 0 \rightarrow (1)$$

$$f(x) \cdot f\left(\frac{3}{x}\right) = 1 \rightarrow (2)$$

from (1), (2)

$$f(x) = 1, \forall x > 0$$